

King distance uses the larger difference

Chess Math

THE IDEA

On an empty board, a king needs at least the larger of the file distance and rank distance to travel between squares. Diagonal steps change both coordinates at once. Enemy attacks can increase that minimum or make a route impossible.

WORKED EXAMPLE

From a1 to d4, the king needs three steps: b2, c3, d4. From a1 to e3, it needs four because the file difference is four. The exact path may vary, but no legal path can beat the geometric minimum.

TRY IT BEFORE READING ON

What is the empty-board distance from b2 to f5?

ANSWER AND REASONING

Four king moves: the file difference is four and the rank difference is three. Use this lower bound when judging whether a king can join a pawn race.

TAKE THIS INTO YOUR NEXT GAME

Calculate the geometric minimum, then add obstacles.